

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

Evaluating generative models hinges on human preferences, because standard metrics often fail to reflect them; yet human evaluations are costly and noisy, as participants vary in expertise, attention, and diligence. Pairwise comparisons improve consistency, but aggregating them into overall quality scores is hard: existing Bradley-Terry-based methods either ignore rater variability or rely on gradient-based fitting without convergence guarantees, which limits their robustness and interpretability. This paper closes that gap with **BBQ**, a Bayesian Bradley-Terry model with rater Quality, and we make three contributions. **(i) Model:** we formulate a Bayesian Bradley-Terry variant that explicitly models per-rater reliability, so the aggregator can down-weight or remove unreliable participants rather than trusting every judgment equally. **(ii) Algorithm:** we derive an Expectation-Maximization algorithm with closed-form updates that jointly estimates item skills and rater quality and guarantees monotonic likelihood improvement, unlike the gradient-based alternatives it replaces. **(iii) Evidence:** across eight human-evaluation datasets, including a newly collected inhomogeneous-rater-quality (IHQ) benchmark, we show that BBQ delivers faster convergence, well-calibrated uncertainty estimates, and more robust, interpretable rankings than baseline Bradley-Terry models, especially under noisy or crowdsourced raters. Together, these contributions make human evaluation of generative models more reliable and cost-effective.

1 INTRODUCTION

Evaluating generative models is hard, particularly for large language models (LLMs) and image generators, because standard metrics often fail to reflect human preferences. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, provide only limited insight into perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Human evaluations therefore remain indispensable for establishing meaningful rankings between models, which is why leaderboards and comparative studies of LLMs and Learned Image Compression (LIC) methods lean so heavily on human preference data. The central problem we address is how to turn such preference data into reliable rankings when the human judgments themselves are noisy.

Human evaluations are expensive and time-consuming, so the protocol must be efficient while minimizing subjectivity and noise. Pairwise comparisons, in which participants choose between two items rather than assigning absolute scores, are an effective and practical answer: they are easier for participants and yield more consistent judgments (Zerman et al., 2018; Wang et al., 2023). Collecting every pairwise comparison is infeasible, however, because the number of pairs grows quadratically with the number of items, and from a limited set of comparisons simple statistics such as win rates do not suffice, since a win rate depends on the strength of the items it was measured against. Producing an overall ranking for a leaderboard thus requires aggregating comparisons effectively, which motivates robust and efficient aggregation methods. The Bradley-Terry (BT) model (Bradley & Terry, 1952) is the standard tool for this task, iteratively updating item scores from comparison outcomes across many domains.

Despite its usefulness, the standard Bradley-Terry model leaves two gaps that our work targets directly. *The first gap is a lack of convergence guarantees for the estimators used in practice.* BT

parameters can be fit with iterative scaling (Ford Jr, 1957), gradient descent, or the minorization-maximization (MM) algorithm (Hunter, 2004). MM converges to the global maximum for the basic BT model, whereas gradient-based methods, closely tied to the Elo rating system and widely used in practice, offer no such guarantee (Hunter, 2004); and for extended BT models with factors such as home-field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM may reach only a local optimum (Hunter, 2004). *The second gap is unmodeled rater variability.* Human evaluations are inherently noisy—participants rush, guess, or lose focus—so reliability varies across raters, and ignoring this variability lets unreliable raters distort item scores and destabilize the ranking.

Contributions. We close both gaps with **BBQ**, a Bayesian Bradley–Terry model that jointly estimates item quality and rater reliability. To the best of our knowledge, no prior work combines the BT framework with a Bayesian formulation that explicitly models rater quality *and* provides closed-form EM updates. Concretely, this paper contributes:

- **A Bayesian BT model with explicit rater quality.** We introduce rater-specific parameters that capture how consistent or trustworthy each participant is, letting the model adjust the influence of individual comparisons. Partially reliable raters still contribute useful information, while inattentive or inconsistent ones are down-weighted; the Bayesian formulation additionally addresses epistemic uncertainty and provides regularization (Section 3).
- **A closed-form EM algorithm with convergence guarantees.** Via a latent-variable formulation we derive an Expectation-Maximization algorithm that efficiently estimates item and rater parameters and guarantees monotonic likelihood improvement, directly addressing the first gap left by gradient-based fitting (Section 3).
- **Interpretable, calibrated uncertainty.** By modeling rater quality within a Bayesian framework, BBQ yields uncertainty estimates for item scores that support more interpretable rankings and robust comparisons across studies (Section 4).
- **A new benchmark and broad empirical validation.** We evaluate on multiple human-evaluation datasets and release a newly collected *inhomogeneous rater quality (IHQ)* dataset¹, showing faster convergence, improved robustness to noisy comparisons, and more consistent rankings than standard BT and naive aggregation (Section 4).

Beyond generative-model evaluation, BBQ applies wherever noisy pairwise comparisons must be turned into reliable global rankings, contributing to more cost-efficient, interpretable, and reproducible human studies of AI-generated content.

2 RELATED WORK

We organize prior work around the two gaps our contributions address—convergence guarantees and rater-quality modeling—and position BBQ against each line of work in turn.

Pairwise comparisons for ranking. Human studies often require ranking items, and asking participants to choose between two items rather than score a single one increases sensitivity to subtle differences and reduces response variability (Zerman et al., 2018; Wang et al., 2023). Many machine-learning leaderboards consequently collect pairwise preferences and aggregate them into overall rankings; notable examples are the Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), both of which combine comparisons using variants of the Bradley-Terry (BT) model.

The Bradley-Terry model and its extensions. The Bradley-Terry model (Bradley & Terry, 1952), originally developed to rank sports competitors, provides a probabilistic framework for the likelihood that one item is preferred over another, turning pairwise comparisons into a ranking; this has made it a cornerstone across games, consumer preferences, and beyond. Several extensions broaden its applicability: the Plackett-Luce model generalizes BT from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a distribution over permutations by

¹The IHQ dataset will be released upon acceptance.

108 multiplying successive BT probabilities (Luce et al., 1959), while other variants model home-field
 109 advantage (Agresti, 2010), ties (Rao & Kupper, 1967), or comparisons between groups rather than
 110 individuals (Huang et al., 2006). None of these extensions model the reliability of the rater who
 111 produced each comparison—the second gap our model closes.

112
 113 **Estimation and the convergence-guarantee gap.** The parameters of the basic BT model are
 114 typically inferred by maximum likelihood estimation (MLE). Zermelo (1929) introduced a widely
 115 adopted iterative procedure for these estimates, which Lange et al. (2000) later identified as a spe-
 116 cial case of minorization-maximization (MM) algorithms that optimize surrogate functions to a local
 117 likelihood maximum. Hunter (2004) extended MM to generalized BT models and gave conditions
 118 for convergence to the MLE (Hunter, 2004): for the classical, convex BT model this yields the global
 119 optimum, but for extensions such as home-field advantage, multiple comparisons, or ties, MM may
 120 converge only to a local maximum (Hunter, 2004). This is precisely the convergence-guarantee gap
 121 that motivates our first contribution.

122
 123 **Bayesian BT.** Bayesian formulations of BT incorporate prior knowledge and regularization
 124 (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed
 125 that MM algorithms can be read as Expectation-Maximization (EM) procedures, enabling Bayesian
 126 inference via Gibbs sampling with tractable complete-data likelihoods and guaranteed convergence
 127 of the resulting Markov Chain Monte Carlo methods. The Bayesian perspective smooths posterior
 128 estimates, reduces susceptibility to local maxima, and quantifies uncertainty in item skills—but this
 129 line of work still assumes every comparison is equally reliable.

130
 131 **Annotator-quality models and our position.** The assumption that all comparisons are equally re-
 132 liable breaks down in crowdsourced evaluations, where participant quality varies widely, motivating
 133 models of annotator reliability (Chen et al., 2013). Chen et al. (2013) jointly estimate item quality
 134 and rater reliability with annotator-specific parameters, placing Gaussian priors on both and using
 135 likelihood scaling factors to modulate each annotator’s influence; they perform posterior inference
 136 by gradient-based optimization, which is difficult in high dimensions and, crucially, carries no con-
 137 vergence guarantee. A simplified Elo-based rater model from Google Research follows the same
 138 spirit and is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and
 139 related work (Mentzer et al., 2020; Ballé et al., 2025). Our contribution is positioned directly against
 140 these two baselines. To the best of our knowledge, no prior work combines the Bradley–Terry model
 141 with a Bayesian formulation that explicitly models rater quality *while also* providing closed-form
 142 EM updates. In contrast to Chen et al. (2013), who rely on Gaussian priors and gradient-based op-
 143 timization without convergence guarantees, BBQ leverages conjugate priors and an EM algorithm
 144 with monotonic likelihood improvement; and in contrast to the heuristic Elo-based rater model used
 145 in CLIC (CLIC, 2025), BBQ provides a principled Bayesian treatment with calibrated uncertainty
 146 estimates and interpretable rater-quality parameters.

146 3 METHODOLOGY

147
 148 This section develops our first two contributions: the rater-quality model (Section 3, this subsection)
 149 and the closed-form EM algorithm that fits it (the Thurstonian view and EM updates below). Con-
 150 verting a set of noisy pairwise comparisons into a reliable item ranking is a fundamental problem
 151 in machine learning and statistics, and it is the problem BBQ is built to solve. Consider K items
 152 repeatedly compared in pairs by a set of R raters, and let D denote the data comprising the outcomes
 153 of these comparisons. For two items i and j , Bradley & Terry (1952) suggested the following model:

$$154 \quad P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

155
 156 where $\lambda_k > 0$ is a parameter associated with item $k \in \{1, 2, \dots, K\}$ that represents its skill rating.
 157 This model infers item scores from observed wins and losses in a clear, interpretable way, but it as-
 158 sumes every comparison is equally reliable—an assumption routinely violated in human evaluation,
 159 where participants differ in expertise, attention, and diligence, producing unreliable and inconsistent
 160 judgments. Relaxing this assumption is the core modeling contribution of BBQ. Concretely, we in-
 161 troduce a rater-specific quality parameter $q_r \in [0, 1]$: with probability q_r rater r makes an informed

judgment following the Bradley-Terry model, and with probability $1 - q_r$ the rater guesses randomly, as if flipping a fair coin between the two items. This single parameter is what lets the aggregator down-weight unreliable participants, and it yields the following mixture model for the probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j by rater r , and $w_{r,ij}$ is the number of times rater r ranks item i above item j . This likelihood has no closed-form maximum likelihood estimator, so λ and q cannot be obtained analytically. Here the two options diverge along the convergence-guarantee gap identified in the introduction: gradient descent is applicable but offers no convergence guarantee, whereas the Expectation–Maximization (EM) algorithm avoids learning-rate tuning and guarantees a monotonic increase of the observed-data likelihood and convergence to a stationary point (Dempster et al., 1977). This is why our second contribution builds the estimator on EM, and the next two subsections derive it.

3.1 THURSTONIAN INTERPRETATION

To make the EM algorithm tractable, we first recast the model with latent variables; this Thurstonian view is the technical device that turns the intractable likelihood above into the closed-form updates that constitute our algorithmic contribution. Following Caron & Doucet (2012), we interpret the Bradley-Terry model under a Thurstonian framework (Diaconis, 1988), viewing a comparison between items i and j as a race in which each item has a random arrival time, Y_i and Y_j , respectively. These arrival times are assumed to follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner. This leads directly to the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we introduce latent variables to simplify the complete-data likelihood. First, we define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which denotes whether the c -th comparison of items i and j by rater r follows the Bradley-Terry model. Using these indicators, we define the latent variable $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma-distributed latent variable is exactly what makes the EM updates tractable while still accounting for rater-specific quality, delivering both halves of our second contribution at once.

3.2 EXPECTATION-MAXIMIZATION UPDATES

We now state the algorithmic contribution in full: an EM procedure whose two steps both reduce to closed-form updates. The EM algorithm iteratively finds maximum a posteriori (MAP) estimates of the model parameters λ and q , treating rater quality and the unobserved Thurstonian arrival times as latent variables, and—unlike the gradient-based fitting used by prior rater-quality models—it is guaranteed to converge to a stationary point of the posterior distribution.

First, we specify prior distributions for the parameters. The item skills λ_k are assigned a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is a conjugate prior for the exponential distribution. Each rater’s quality parameter, q_r , is given a Beta prior, $q_r \sim B(\alpha, \beta)$.

The core of the EM algorithm is the iterative maximization of the expected complete-data log-posterior, conditioned on the current parameter estimates (λ^*, q^*). The algorithm proceeds by alternating between two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, a function we denote as Q . This function represents the expected value of the log-posterior of all observed and latent variables, given our observed data and the current parameter estimates from the previous iteration. It is defined as:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , is further broken down into three components.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the parameters by maximizing the Q function, and this is where the payoff of the latent-variable formulation becomes concrete: using the expected values from the E-step, the intractable original optimization collapses into simple closed-form updates. Both updates hinge on a single quantity—the posterior probability that a given comparison from rater r follows the Bradley-Terry model. Denoted γ , it is the weight of a rater’s judgment according to how well it aligns with the model’s current predictions, and it is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . Intuitively, γ is our confidence that a comparison was meaningful rather than random given the current estimates, so higher γ marks more trustworthy comparisons—the concrete mechanism by which the model realizes its rater-quality contribution. The rater-quality update q_r is a regularized weighted average: the numerator accumulates the “effective number of wins” for that rater, each weighted by the posterior probability γ that the judgment was meaningful rather than random, combined with the Beta-prior hyperparameters; the denominator normalizes by the total number of comparisons and the prior parameters. The update thus increases a rater’s quality score when their judgments frequently agree with the model, and is given by:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The item-skill update λ_i balances two quantities in a ratio. The numerator sums the “effective wins” for item i across all raters, again weighting each win by γ , so it captures the total positive evidence for item i ; the denominator accounts for all comparisons item i was involved in and acts as a normalizer. Both terms are regularized by the Gamma-prior hyperparameters. Critically, the same γ weights that quantify rater quality flow directly into this skill update, so unreliable raters automatically contribute less evidence. The update is given by:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The full derivation of these closed-form EM updates—the technical core of our algorithmic contribution—is provided in Appendix B.

4 EXPERIMENTAL RESULTS

Our experiments are organized to substantiate the contributions claimed above: robustness and ranking accuracy under noisy raters (Table 1), the effect of the convergence guarantee under sparse data (Figure 1), the interpretability of learned rater quality (Figure 2), computational efficiency from the closed-form updates (Figure 3), and calibration of the uncertainty estimates (Figure 6). Throughout, we compare our Bayesian Bradley-Terry model with rater Quality (**BBQ**) against the two baselines it is designed to improve on: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012), which is Bayesian but ignores rater quality, and a gradient-descent-based BT model with rater quality (Crowd-BT) (Chen et al., 2013), which models rater quality but lacks convergence guarantees. For Crowd-BT, we use the Google Research implementation (Google, 2025) employed by both CLIC (2025) and Ballé et al. (2025), and we report all hyperparameters in Section C.

We span a deliberately diverse set of Bradley-Terry (BT) benchmarks: human-preference datasets for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with our newly collected inhomogeneous rater quality (IHQ) dataset from the CLIC2024 image test set (CLIC, 2025). To isolate the effect of rater quality—the very factor BBQ is built to handle—we partition IHQ into screened and unscreened subsets, letting us examine how high-quality, low-quality, and mixed comparisons affect each model. Details of the IHQ dataset creation are given in Section E, and a summary of all datasets—comparisons, raters, and items—appears in Section D and Table 2.

Ground-truth rankings are unavailable for real datasets, so we establish an evaluation protocol in three steps. First, on simulated data all methods recover the same ordering as the number of samples grows, validating the setup. Second, for crowd-sourced datasets we approximate the ground truth by the ranking obtained on the whole dataset, and we sanity-check this approximation via the top-1 item: for every dataset except IHQ-unscreened, all three models agree on the top-ranked item. This last case already previews our robustness contribution: on IHQ-unscreened, Crowd-BT and Bayes-BT both fail to identify the reference image as the highest-quality item, whereas BBQ succeeds.

Third, because a reliable aggregation method should reproduce the same ordering when the study is repeated, we quantify this stability by bootstrapping, which estimates the variability of the rankings (details in Section F). One caveat applies: the WD and HiFiC studies actively selected comparison pairs—HiFiC by binary search, WD by maximum information gain—so their bootstrapping results should be read with caution.

We report two complementary metrics that map onto the two ways rankings are used in practice. Top-1 agreement—the fraction of bootstrapped samples identifying the same best item as the full dataset—matters when only the single best model is of interest. Kendall’s Tau (τ), the ordinal correlation between rankings (Kendall, 1938) where higher is more stable, assesses the stability of the ranking as a whole.

4.1 PERFORMANCE ACROSS DATASETS

This subsection tests the robustness and ranking-accuracy contribution head-on, comparing all three models on accuracy and stability; Table 1 summarizes Top-1 agreement and Kendall’s Tau. As a general trend, datasets with more comparisons per model (MT-Bench, WD, HiFiC) are easier, though HUMAINE breaks this pattern—a reminder that factors beyond raw comparison count, such as rater consistency and diversity, shape performance and motivate modeling rater quality directly.

The headline result is that BBQ is the most stable and robust method across datasets. It identifies the top-performing item most frequently, sharing the best Top-1 agreement on three datasets, and it recovers the overall ranking most accurately on five of the eight datasets, ranking second on the remaining three.

Where BBQ places second in Kendall’s Tau, the reason is instructive rather than adverse: all three such datasets have high-quality, homogeneous rater sets. On MT-Bench every model perfectly re-

Table 1: Evidence for the robustness contribution: performance of BT-based aggregation methods across eight datasets (top: Top-1 agreement [%]; bottom: Kendall’s Tau; best in **bold**, second underlined). Supporting our central claim, BBQ most frequently identifies the top-performing item, recovers the best overall ranking on more than half of the datasets, and ranks second on the rest—with its largest margins on the noisy IHQ-all and IHQ-unscreened subsets it is designed for.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	0.9245	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

covers the top item; the WD dataset of Ballé et al. (2025) has only five raters, likely the authors themselves and thus exceptionally careful; and IHQ-screened explicitly filtered raters for quality. When rater quality is uniformly high, there is little variability for BBQ to exploit, so its advantage over simpler models naturally shrinks—exactly the regime in which our contribution is expected to matter least.

The complementary regime is where the contribution pays off, and the IHQ screened/unscreened split makes the pattern clear. BBQ substantially outperforms Crowd-BT and Bayes-BT whenever low-quality raters are present, as in IHQ-all and IHQ-unscreened. All three models attain their best Top-1 accuracy on the screened subset, confirming the value of rater selection, but the methods diverge sharply without it: although Crowd-BT also models rater quality, its Top-1 accuracy drops noticeably on the full dataset—likely because crowdsourced raters contribute fewer than 40 comparisons each, leaving it vulnerable to noise—whereas BBQ holds up with only a minor decrease, directly demonstrating robustness to low-quality raters.

This robustness matters because uniformly high rater quality is hard to achieve in practice: large-scale crowdsourcing introduces variability, screening is costly, and subjective factors affect even diligent annotators. In exactly these conditions, BBQ delivers on its promise—leveraging partially reliable raters while suppressing the impact of noisy ones.

4.2 SCALING WITH RATERS AND COMPARISONS

Having established overall accuracy, we now isolate the practical value of the convergence guarantee by varying data density. Table 1 already showed that more comparisons per model help, and the comparison budget can grow in two ways: more raters, or more comparisons per rater. Figure 1 disentangles these two factors—the left column varies the number of raters (comparisons per rater fixed at the maximum) and the right column varies comparisons per rater (raters fixed at the maximum). The models separate clearly in Top-1 accuracy while Kendall’s τ stays similar, and the decisive observation is at the sparse end: Crowd-BT fails to converge with only one or two raters, which is exactly the advantage our EM formulation buys over gradient descent.

Crowd-BT and BBQ perform comparably when raters or comparisons per rater are few, since BBQ needs both multiple raters and multiple comparisons per rater to tell rater qualities apart; as either grows, BBQ increasingly outperforms Bayes-BT. Bayes-BT is the weakest overall but can edge out Crowd-BT in the sparsest regimes, precisely because Crowd-BT cannot reliably estimate rater quality when data per rater is scarce—reinforcing that a guaranteed-convergent estimator is what makes rater-quality modeling pay off in practice.

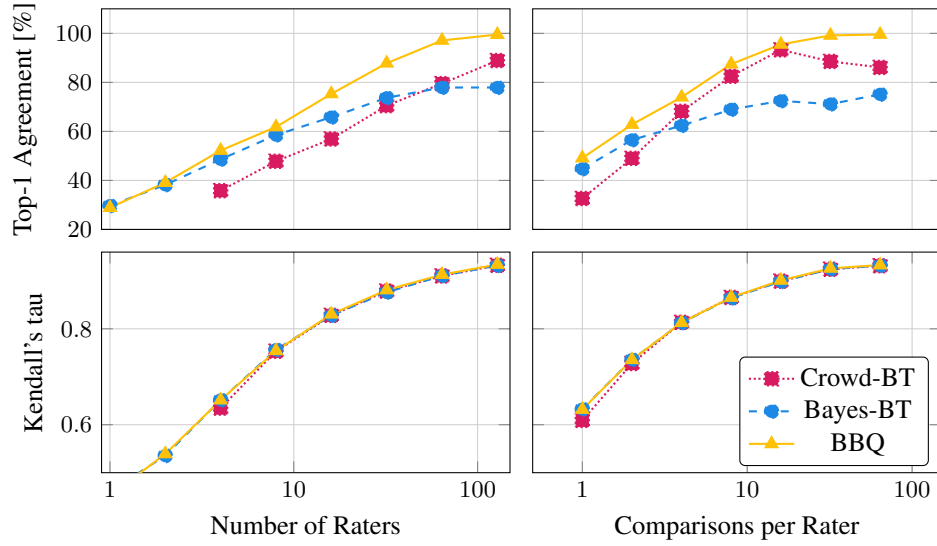


Figure 1: Evidence for the convergence-guarantee contribution: scaling behavior of the BT variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset. **Left:** performance vs. number of raters; **Right:** performance vs. number of comparisons per rater. Both Top-1 agreement and Kendall’s τ improve with more raters or comparisons, and while Top-1 agreement separates the models, Kendall’s τ stays similar across them. The key takeaway is at the sparse end: Crowd-BT fails to converge with very few raters, whereas BBQ’s EM algorithm does not—the practical benefit of our convergence guarantee. Crowd-BT and BBQ track each other under sparse data, but BBQ pulls ahead of Bayes-BT as raters or comparisons grow.

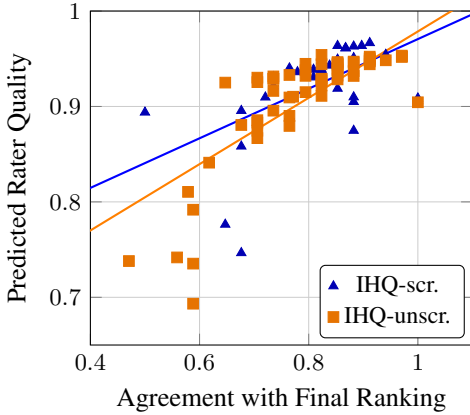


Figure 2: Evidence for the interpretability contribution: predicted rater quality (y-axis) versus each rater’s agreement with the final ranking (x-axis) on the IHQ datasets, one point per rater (triangles: filtered; squares: unfiltered). The positive trend shows that BBQ’s learned quality parameters recover a genuinely interpretable notion of reliability rather than an opaque weight.

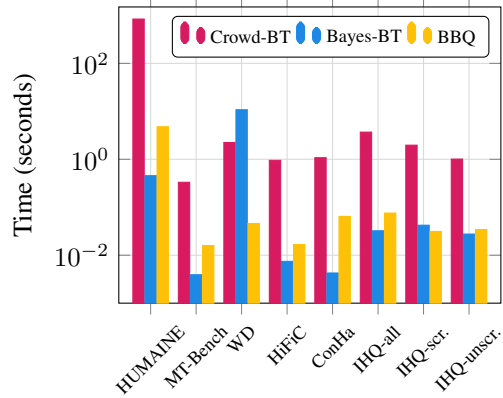


Figure 3: Evidence for the efficiency benefit of closed-form EM updates: average computation time in seconds (log-scale) per bootstrapped sample across the eight datasets. Whereas Crowd-BT can be costly on some datasets, BBQ stays fast on every one, making repeated bootstrapping practical.

4.3 RATER QUALITY

We next ask whether BBQ’s rater-quality parameters are interpretable, the third of our claimed contributions. Figure 2 relates each rater’s agreement with the final ranking to the quality BBQ assigns them, and the answer is affirmative: the two are clearly positively correlated, so raters who align with the consensus receive higher predicted quality. The Pearson correlation is $r = 0.551$ over 50 raters on the filtered dataset and a stronger $r = 0.724$ over 62 raters on the unfiltered dataset. As expected, the unscreened set contains several raters with both low agreement and low predicted quality, and some even fall below chance (50% agreement), systematically disagreeing with the majority. This is precisely the failure mode that motivates modeling rater quality: BBQ identifies these unreliable raters, assigns them low quality scores, and thereby curbs their influence on the final ranking—turning the interpretability of the parameters into concrete noise mitigation.

4.4 COMPUTATIONAL EFFICIENCY

A final practical consequence of the closed-form EM updates is speed, which we quantify here. Figure 3 reports the average time each method needs to process one bootstrapped sample per dataset—a direct read on real-world feasibility. BBQ converges within a few seconds on every dataset. Crowd-BT is slower across the board and especially costly on comparison-heavy datasets such as HUMANMAINE, where a single run takes around 15 minutes; Bayes-BT is slow only on WD and otherwise marginally faster than BBQ. BBQ’s consistent speed is a direct dividend of the closed-form EM updates derived in Section 3, which converge rapidly even on large datasets. The comparison is in fact conservative: Crowd-BT is a highly optimized C implementation (Kernighan & Ritchie, 1988), whereas BBQ runs in plain NumPy with no special optimization, so a compiled or vectorized version could be faster still. Taken together with the accuracy and stability results above, this makes BBQ not just more robust but also practical for large-scale preference studies and repeated bootstrapping—a compelling package for real-world deployment.

4.5 CONFIDENCE INTERVALS

The last contribution to validate is calibrated uncertainty, which underpins the interpretable, significance-aware rankings BBQ promises. In a controlled simulation, raters compare equally skilled items (H_0), and each model estimates skills and builds uncertainty intervals to test for significant differences (Section G). The resulting Type I error rates appear in Figure 6; at the 99% confidence level the expected rate is 1%. BBQ and Crowd-BT both land close to this target (slightly below), while Bayes-BT is markedly conservative at $\sim 0.1\%$. BBQ therefore delivers well-calibrated uncertainty, whereas Bayes-BT understates false positives—confirming that BBQ’s error bars can be trusted for significance testing between items.

5 LIMITATIONS

We close by delimiting where our contributions do and do not apply. First, the scope of the rater-quality contribution is conditional: BBQ’s advantage diminishes when all raters are uniformly reliable, because there is little variability left to model in such homogeneous datasets. Since guaranteeing uniformly high quality is itself costly and often infeasible at scale, this locates BBQ’s value in realistic crowdsourced settings rather than in carefully curated ones. Second, the model contribution is currently limited to pairwise comparisons, whereas many studies collect ratings, rankings, or multi-way inputs; extending BBQ to these feedback types would broaden its reach. Third, on the evidence side, our stability estimates rest on a finite number of bootstrapped comparisons, so larger-scale studies would further validate robustness, and the model assumes independent comparisons, ignoring contextual or order effects that can influence human judgments. Finally, although the closed-form updates make BBQ fast, very large numbers of items or raters could still strain a plain implementation absent compiled or vectorized code.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability, and we delivered on the three contributions set out in the introduction. On the modeling side, explicit rater-quality parameters let BBQ down-weight or remove unreliable participants, producing more stable and accurate rankings. On the algorithmic side, the closed-form Expectation-Maximization (EM) algorithm estimates item skills and rater quality simultaneously with rapid convergence and monotonic likelihood improvement, resolving the convergence-guarantee gap of gradient-based approaches. On the empirical side, across eight diverse datasets—including our new IHQ benchmark—BBQ attained high Top-1 agreement and strong Kendall’s Tau, with its clearest gains exactly where the contribution is meant to help: noisy, crowdsourced settings where many raters are non-experts, vary in diligence, or contribute few comparisons. These results confirm that modeling rater quality is most crucial when evaluation quality is heterogeneous or partially unreliable. Two further payoffs of the Bayesian formulation round out the evidence: BBQ provides principled uncertainty estimates for item scores whose error bars we showed to be well-calibrated and usable for significance testing, and its predicted rater quality aligns with agreement to the final ranking, validating that it identifies reliable evaluators.

Taken together, these contributions advance human-evaluation methodology with a practical, interpretable, and generalizable way to aggregate noisy pairwise comparisons, a concrete step toward more cost-effective, interpretable, and reproducible human studies of AI-generated content.

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A USAGE OF LLMs

During the preparation of this paper, we utilized large language models (LLMs) as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, refining phrasing, and condensing text to enhance clarity and readability. Additionally, ChatGPT, Claude, and Cursor were used for code analysis, completion, and the generation of visualizations to aid our development workflow. These tools were employed solely as auxiliary aids, while all core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items. Each rater r has quality q_r , meaning that with probability q_r they follow the Bradley–Terry model, and with probability $1 - q_r$ they choose randomly. The following is a standard EM (Dempster et al., 1977) derivation that incorporates these rater-specific reliabilities into the Bayesian estimation framework.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley–Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley–Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley–Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley–Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function w.r.t. the parameters (λ, q) , holding the expectations computed in the E-step fixed.

Update for q_r The update for each rater quality q_r is obtained by maximizing Q with respect to q_r (including the Beta prior).

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i (including the Gamma prior).

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with the default hyperparameters.

The only hyperparameters in the Bayes-BT and BBQ models are the prior distribution parameters and the stopping thresholds. For both Bayes-BT and BBQ, we consider the model converged when no ELO score changes by more than 1 between two iterations. The ELO score can be calculated from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set the ELO.SCALE.FACTOR to 400.

We chose a gamma prior with shape 5 and rate 0.1 for the skill parameters in both Bayes-BT and BBQ. For BBQ, the beta prior on the rater quality has $\alpha = 10$ and $\beta = 2$.

D DATASETS

Our evaluation deliberately spans language and image domains so that the robustness contribution is tested across scale and rater expertise; Table 2 gives an overview. For natural language we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which collects crowd-worker judgments on multi-turn dialogues. For image compression we use three datasets that differ sharply in scale, rater count, and expertise—WD (Ballé et al., 2025), densely expert-labeled with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), evaluating learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), targeting conditional generative models—providing a broad testbed for robustness.

The centerpiece of our data contribution is the inhomogeneous rater quality (**IHQ**) dataset, collected in a user study on the CLIC2024 (CLIC, 2025) data via the *Mabyduck* platform. IHQ comprises two-alternative-forced-choice (2AFC) judgments across 28 generative image-compression models and, uniquely, ships with two subsets designed to expose rater-quality effects: (i) **screened**, whose raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, which retains all raters. The screened subset represents higher-quality raters, while the unscreened subset reflects the noisy conditions typical of large-scale human evaluation. Together, these datasets let us probe both large, relatively clean benchmarks and smaller, noisier settings—the regime where modeling rater reliability is most critical and where BBQ’s contribution is meant to show.

Table 2: Summary of the eight datasets used in our experiments. The IHQ dataset is our own contribution, collected on the Mabyduck platform (2AFC and 3AFC settings), and is split into screened and unscreened subsets: screened includes only raters who passed pre-screening tests for attention and display quality (higher reliability), while unscreened includes all raters. The wide range of comparison and rater counts is what lets these datasets stress-test robustness.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

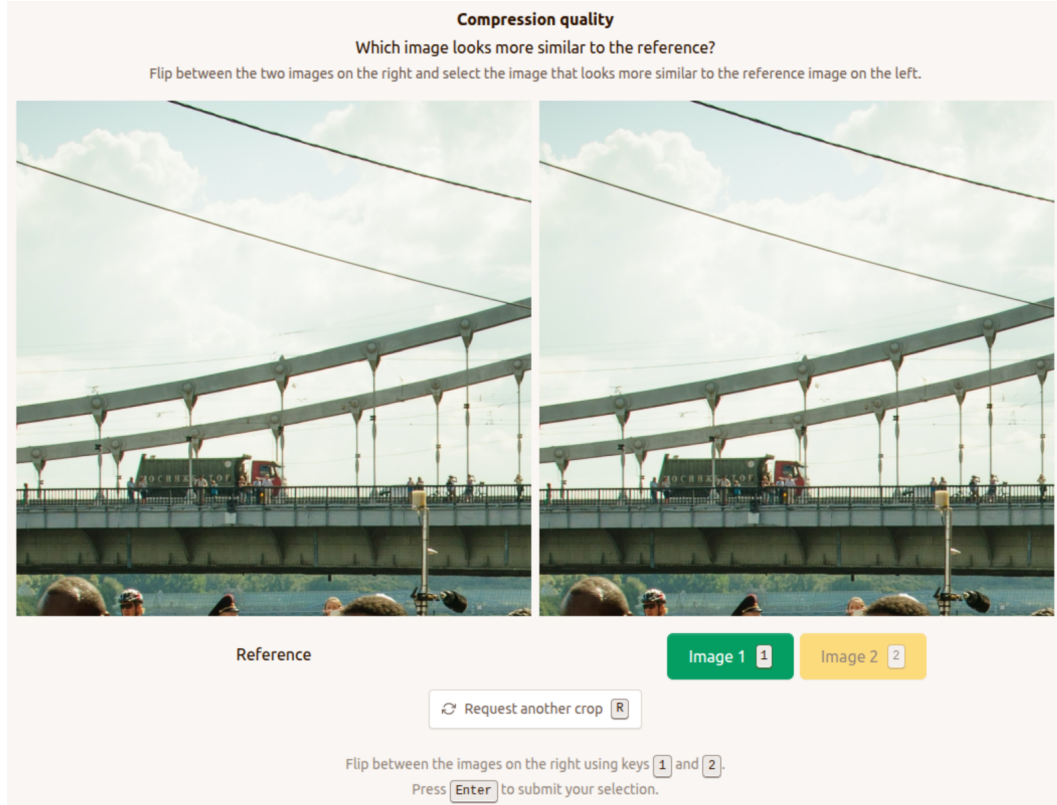


Figure 4: The Mabyduck user-study platform used to collect the IHQ pairwise comparisons: a reference image on the left, and the two compressed images the rater chooses between on the right.

E USER STUDY PLATFORM

This appendix documents how the IHQ dataset contributed by this paper was collected. All pairwise comparisons on the CLIC2024 (CLIC, 2025) data were gathered with the Mabyduck platform (Ltd., 2025), shown in Figure 4. The task asks, “Which image looks more similar to the reference image?”, ties are disallowed, and all pairs were drawn uniformly at random.

The screened/unscreened split is what makes IHQ useful for studying rater quality: screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and com-

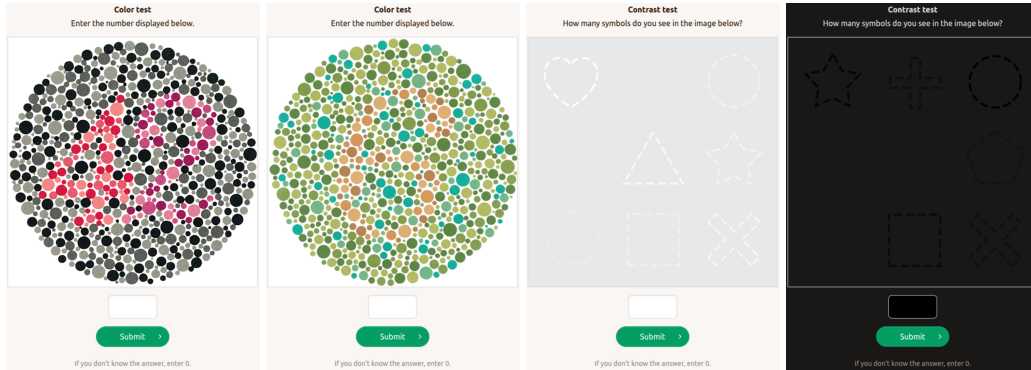


Figure 5: Four example images from the rater pre-screening that defines the screened IHQ subset. The first two are color-blindness tests (identify the number in each pattern); the last two are low-contrast shape-detection tests—a light-gray shape on white and a dark-gray shape on black.

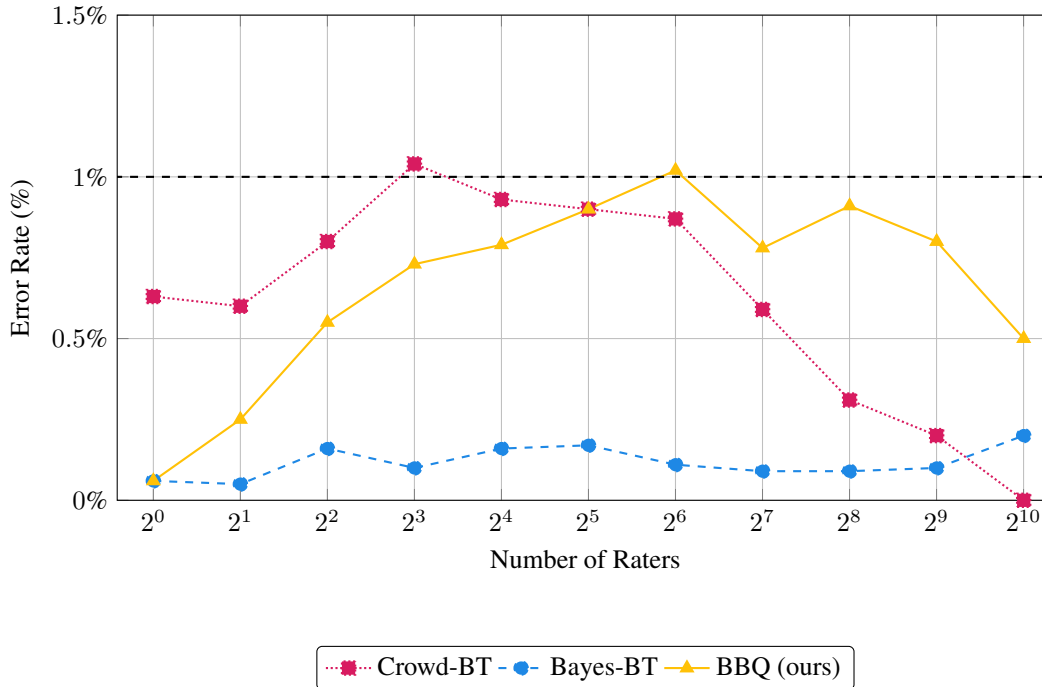


Figure 6: Calibration evidence: Type I error rate versus the number of raters, with the dashed line marking the expected 1% rate at the 99% confidence level. BBQ and Crowd-BT track this target closely, whereas Bayes-BT stays well below it.

pleted training comparisons in which they selected the higher-quality image to reinforce the evaluation criteria; unscreened raters did not.

Figure 5 shows four example pre-screening images. The first two are standard color-blindness tests in which raters must identify the number in each pattern; the last two are shape-detection tests probing sensitivity to low-contrast objects—one light-gray shape on a white background, one dark-gray shape on a black background. These checks ensure that only raters with adequate visual acuity enter the screened subset, which is what gives the screened/unscreened contrast its meaning.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT model rater quality explicitly, we bootstrap over raters rather than individual comparisons; this preserves each rater’s comparison distribution and yields a fair assessment of stability. We use 10,000 bootstrap resamples of raters for every dataset, with one exception: on HUMAINE a single Crowd-BT iteration takes about 15 minutes (Section 4.4), so we reduce to 1,000 resamples there to keep the experiment tractable.

G UNCERTAINTY ESTIMATION DETAILS

This appendix details the uncertainty machinery behind the calibration contribution: beyond ranking items, estimating uncertainty is what lets us judge whether an observed difference is statistically significant. The Bayesian models (Bayes-BT and BBQ) quantify uncertainty directly from the posterior over item skills: each skill carries a Gamma prior updated by the observed comparisons, and the resulting posterior credible intervals are mapped to the Elo scale for cross-item comparison. Crowd-BT, being non-Bayesian, instead uses a frequentist approximation—a second-order Taylor expansion around the maximum likelihood estimate, inverting the Hessian of the log-likelihood to obtain a covariance matrix whose diagonal gives per-item variances. These variances become 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to the 99% level; narrower likelihood peaks give smaller intervals (higher certainty) and flatter peaks give larger ones (greater uncertainty).

To test how well each model estimates uncertainty, we run a controlled simulation. Each trial pits two items of equal skill (50–50 win probability) against each other, with pairwise data generated by coin flips, and simulates multiple raters before applying each model to estimate Elo scores and their uncertainty. The null hypothesis H_0 is that the two items are equally strong; we compute each item’s 99% interval and count a trial as an incorrect rejection of H_0 whenever the intervals fail to overlap, signaling a significant difference where none exists. The metric of interest is thus the observed Type I error rate—how often a model wrongly declares the items different. Over 10,000 trials with 50 comparisons per rater and a varying number of raters, we compare each model’s empirical error rate against the theoretical 1%. The verdict, shown in Figure 6, supports our calibration claim: Crowd-BT and BBQ sit close to the expected 1%, so their uncertainty estimates are well-calibrated, whereas Bayes-BT is consistently over-conservative, indicating that its credible intervals are less suited to significance testing.